

Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

+91 88008-12254 | garvarora0205@gmail.com | linkedin.com/in/gaminization | github.com/gaminization | garvarora.vercel.app

Education

Vellore Institute of Technology (VIT)

Vellore, India

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

Aug 2023 – Jul 2027

Gems International School (CBSE)

Gurugram, India

Class XII: 76.4% | Class X: 89.3%

Nov 2016 – Apr 2023

Work Experience

LG Soft India (LGSi) – HS/ES India Lab

Noida, India

Living Solution Control Development Intern

Jun 2026 – Jul 2026

- Wrote C/C++ control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; rated “Technically Competent” by LGSi HR and engineering leads.

Samsung PRISM

Remote

Data Intelligence Agent Developer (Internship)

Apr 2026 – Oct 2026

- Built a multi-agent data intelligence system using LangGraph, FastAPI, Redis, PostgreSQL, and Qdrant to discover datasets and execute a 20-stage analysis pipeline; integrated spaCy, Presidio for PII redaction, and MemorySaver human-in-the-loop checkpoints.

Wissen Baum Engineering Solutions LLP

Pune, India

Software Automation Intern

May 2025 – Jul 2025

- Put together a Python test automation framework with Gherkin/Behave, pytest, and Playwright, parallelizing GitLab CI jobs to cut pipeline time from ~10 to ~6 minutes (40% faster) and reduce manual testing by 80%+.

SEDS India (SEDS VIT)

Vellore, India

R&D Lead & Autonomous Systems Developer

Apr 2024 – Jul 2026

- Set up ROS2 Nav2 with RTAB-Map 3D Visual SLAM on an Intel RealSense D455, generating real-time elevation maps and cutting manual rover teleoperation time by 70% during competition.
- Rewrote the CV pipeline using CUDA GPU kernels, YOLOv8, and ArUco/AprilTag tracking, tuning Linux V4L2 for zero-copy buffers across 4 simultaneous camera streams to drop latency by ~60% and boost throughput 4x.
- Flashed Micro-ROS onto ESP32 microcontrollers for sensor polling, processing wheel encoder and IMU telemetry through an Extended Kalman Filter (EKF) to cut odometry drift by ~40% (sub-10 cm accuracy).
- Built a PyQt5 mission control dashboard with real-time telemetry visualization, live parameter tuning, and an FSM handling autonomous mission execution.

Projects

teleop-cursor | *ROS 2, Python, rclpy, geometry_msgs/Twist* | pypi.org/project/teleop-cursor

Feb 2026 – Apr 2026

- Developed zero-hardware ROS 2 mouse cursor teleoperation node that converts screen movement into geometry_msgs/Twist velocity commands at 10 Hz for TurtleBot3 Gazebo simulation.

HavaiOS – Preemptive Bare-Metal RTOS Kernel | *C, ARM Assembly, ARM Cortex-M4*

Jan 2026 – Mar 2026

- Wrote a preemptive bare-metal RTOS kernel for ARM Cortex-M4 in C and Assembly featuring 1 ms tick scheduling, context switching in <50 instructions, mutex/semaphore primitives, and an isolated HAL.

Captivity CLI | *Rust, Python, Linux D-Bus, systemd, Socket IPC* | pypi.org/project/captivity-cli

Nov 2025 – Jan 2026

- Built autonomous WiFi captive portal login daemon featuring <50ms HTTP 204 probing, D-Bus NetworkManager monitoring, systemd background integration, keyring credential encryption, and Python-Rust socket IPC.

3D Reconstruction – IMU-Enhanced KinectFusion | *C++, OpenCV, Intel RealSense SDK*

Aug 2025 – Oct 2025

- Fused RealSense D455 depth data with IMU readings through a complementary filter to stabilize KinectFusion TSDF reconstruction, dropping tracking failures by ~50% during fast camera motion.

Patents

Patent Published (App IN202641072249 A1): Autonomous Radar Survivor Detection

Jun 2026

- Fused 24GHz FMCW mmWave radar, ultrasonic sensors, and IMU dead-reckoning on dual-core ESP32; created confidence grid decay formula $C = \max(0, C - \beta \Delta t)$ for survivor breathing detection under rubble.

Achievements & Awards

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate).

International Rover Challenge (IRC): Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT).

Certifications

AWS Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner

Oracle Cloud Certifications: Certified Data Science Professional | Certified Generative AI Professional

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS